Computer Science - Graduate Level Assessment 21 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Local caching of files is common in distributed file systems, but it has the disadvantage that
   (a) Temporary inconsistencies among views of a file by different machines can result
   (b) The file system is likely to be corrupted when a computer crashes
   (c) A much higher amount of network traffic results
   (d) Caching makes file migration impossible

2. Of the following, which gives the best upper bound for the value of f(N) where f is a solution to the recurrence f(2 N + 1) = f(2 N) = f(N) + log N for N >= 1, with f(1) = 0?
   (a) O(log N)
   (b) O(N log N)
   (c) O(log N) + 0(1)
   (d) O((log $N)^2$)

3. Applications developed by programming languages like ______ and _________ have this common buffer-overflow error.
   (a) C, Ruby
   (b) Python, Ruby
   (c) C, C++
   (d) Tcl, C#

4. The stack is memory for storing
   (a) Local variables
   (b) Program code
   (c) Dynamically linked libraries
   (d) Global variables

5. Of the following problems concerning a given undirected graph G, which is currently known to be solvable in polynomial time?
   (a) Finding a longest simple cycle in G
   (b) Finding a shortest cycle in G
   (c) Finding ALL spanning trees of G
   (d) Finding a largest clique in G

True / False (10 marks)

Answer True or False

6. True or False: A session symmetric key can be reused multiple times across different sessions without compromising security.
7. True or False: Message integrity ensures that data arrives at the receiver exactly as it was sent, without alteration during transmission.
8. True or False: UNIX-based operating systems have historically been vulnerable to buffer-overflow attacks due to their handling of memory management.
9. True or False: The AH Protocol provides source authentication and data integrity, but it does not provide privacy for the transmitted data.
10. True or False: Church's thesis was first proven by Alan Turing, establishing a formal basis for computable functions.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to remove particular data type elements from the given tuple.
12. Write a function to separate and print the numbers and their position of a given string.

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